DELHI PUBLIC SCHOOL NEWTOWN
SESSION 2023-24
HALF HEARLY EXAMINATION

CLASS: X **FULL MARKS: 80**
SUBJECT: PHYSICS (SET A) **TIME: 2 HOURS**

Candidates are allowed additional 15 minutes for only reading the paper.
They must NOT start writing during this time.
Section A is compulsory. Attempt any four questions from Section B
This paper consists of six printed pages

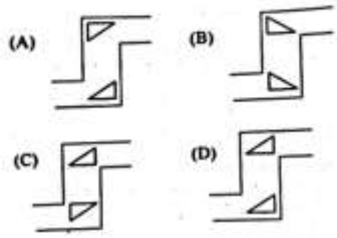
SECTION A

(Attempt all questions from this section)

1. **[15 × 1 = 15]**

(i) The refractive index of diamond is 2.4. It means that:
- **(a) the speed of light in vacuum is equal to 1/2.4 times the speed of light in diamond.**
- **(b) the speed of light in the diamond is 2.4 times the speed of light in a vacuum**
- **(c) the speed of light in a vacuum is 2.4 times the speed of light in the diamond**
- **(d) the wavelength of light in diamond is 2.4 times the wavelength of light in vacuum**

(ii) Which of the following diagrams show the correct use of prisms in the periscope?



(a) A **(b) B** **(c) C** **(d) D**

(iii) Two sound waves (S_1 and S_2) produced from two different sources are of same amplitude and are having frequencies 256 Hz and 512 Hz respectively. Then,
- **(a) S_2 is louder than S_1**
- **(b) S_1 is shriller than S_2**
- **(c) S_1 is called infrasonic and S_2 is called ultrasonic.**
- **(d) S_2 is shriller than S_1**

(iv) When an irregular body is freely suspended from a point, its centre of gravity at balanced position lies
- **(a) vertically above the point of suspension.**
- **(b) geometric centre of the irregular body.**
- **(c) at the base of the irregular body.**
- **(d) vertically below the point of suspension**

(v) If a block and tackle system with effort applied in convenient direction has 3 movable pulleys, then its velocity ratio
(a) is either 6 or 7 **(b) should be 6** **(c) should be 7** **(d) is 3**

(vi) If a light body A of mass 'm' and a heavy body B of mass 'M' have the same kinetic energy 'K', then the heavy body B will have:
(a) more momentum than the light body A.
(b) less momentum than the light body A.
(c) same momentum as that of A.
(d) none of the above is true.

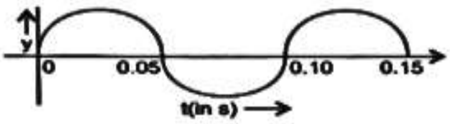
(vii) The visible colour that has the highest critical angle is
(a) red (b) green (c) violet (d) yellow

(viii) A particle of mass m moves from rest under the action of a constant force F which acts for 2s. The maximum power attained by the particle is
(a) $(2F^2)/m$ (b) $(F^2)/m$ (c) $m/2F^2$ (d) 2Fm

(ix) Select the incorrect statement from the following
(a) The efficiency of a real machine is always less than 100%
(b) The mechanical advantage of a machine can be less than 1
(c) A machine can be used as a speed multiplier
(d) The mechanical advantage of a machine can be greater than its velocity ratio.

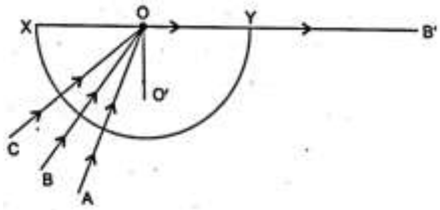
(x) Magnification produced by concave lens is always:
(a) more than 1 (b) equal to 1 (c) less than 1 (d) none of the options

(xi) The diagram shows displacement-time graph of a wave travelling in a string with velocity 20 ms^{-1}.



The value of wavelength is
(a) 1 m (b) 1.5 m (c) 2m (d) 2.5m

(xii) Three rays of light A, B and C of the same colour are incident on the slab and strikes on the edge XY at the point O. The light ray B suffers refraction along OB'.



Name the phenomenon that the ray A suffers
(a) refraction (b) reflection (c) total internal reflection (d) none of the options

(xiii) When a pair of scissors is used to cut the clothes, its blades move longer on cloth by a small displacement of its handles. In this case, effort is
(a) equal to the load (b) less than the load
(c) more than the load (d) information is insufficient to conclude

(xiv) The source of ultraviolet light is
(a) Electric bulb (b) Red hot iron ball (c) Sodium vapour lamp (d) Carbon arc-lamp

(xv) Assertion (A) : When a convex lens is dipped in water, its focal length increases.
Reason (R) : When a convex lens is dipped in water, the extent of bending of light increases.
(a) Both A and R are true, and R is the correct explanation of A.
(b) Both A and R are true, and R is not the correct explanation of A.
(c) A is true, but R is false.
(d) A is false, but R is true.

Question 2

(i) (a) Which lens has more power: a thick or a thin lens?
(b) The power of a lens is -4D. Determine its focal length.
(c) Mention the relationship between critical angle and refractive index of any medium. [3]

(ii) Classify the following into lever of different class:
a) Bottle opener b) A long handled oar [2]

(iii) Two strings of diameter 1 cm and 1.5 cm are set into vibration. Which of them will produce sound of higher pitch? Give reason to your answer. [2]

(iv) A coin placed at the bottom of a beaker appears to be raised by 4.0 cm. If the refractive index of water is 4/3, find the depth of the water in the beaker. [2]

(v) A boy uses blue colour of light to find the refractive index of glass. He then repeats the experiment using red colour of light. Will the refractive index be same or different in the two cases? Give a reason to support your answer. [2]

(vi) (a) What is the ratio of the velocities of light of wavelengths 4000Å and 8000Å in vacuum?
(b) Which of the above wavelengths has a higher frequency? [2]

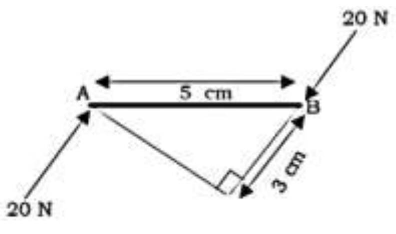
(vii) (a) Name the SI unit of physical quantity obtained by the formula 2K/ V^2 where K: kinetic energy, V: Linear velocity.
(a) How is uniform circular motion different from uniform linear motion? [2]

Question 3

(i) A simple pendulum is oscillating in a medium. Draw the displacement versus time graph representing the motion. [2]

(ii) If kinetic energy of a moving body is 40J, then what will be its kinetic energy when its velocity is halved [Relevant calculation is to be shown]? [2]

(iii) The diagram given below shows two parallel forces acting on rod AB. Calculate the total moment acting on the rod AB in S.I. unit. [2]



(iv) A lens produces a virtual image formed between the object and the lens.
a) Mention the type the lens.
b) Write any one use of such lenses. [2]

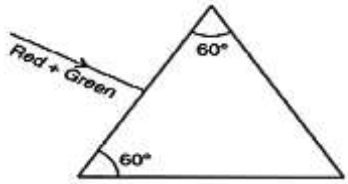
(v) A ball of mass 15 g is thrown vertically upward with initial velocity 20 ms^{-1}. At what height above the ground the ball would have equal potential and kinetic energy? [Take g = 10 ms^{-2}]. [2]

SECTION B

(Attempt any four questions)

Question 4 **[3+3+4]**

(i) The diagram shows a beam of light (red + green) incident normally on an equilateral triangular prism. If the critical angle for the material of prism is 60° for the light of red colour, complete the diagram showing the path of light of each colour emerging out of the prism. Mark the angles in the diagram where ever necessary.



(ii) A lens of focal length 15 cm forms an image on the screen of size three times that of the object.

(a) What kind of lens is used?

(b) Find the position of object.

(iii) A 50 cm uniform ruler is freely pivoted at 15 cm mark which balances horizontally when an object of weight 40 gf is hung from the 2 cm mark.

(a) Draw a diagram of the arrangement.

(b) Calculate the weight of the ruler.

Question 5 **[3+3+4]**

(i) A pendulum has a frequency of 4 vibrations per second. An observer starts the pendulum and fires a gun simultaneously. He hears the echo from a cliff after 6 vibrations of the pendulum. If the velocity of sound in air is 340 ms^{-1}, find the distance between the cliff and the observer.

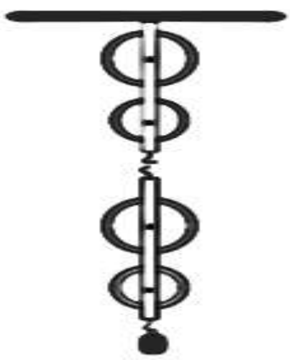
(ii) If a man raises a box of 50 kg to a height of 2 m in 2 min, while another man raises the same box to the same height in 5 min. Compare

(a) work done (b) power developed by them.

(iii) The diagram below shows a block and tackle system.

(a) Copy and redraw the diagram showing the correct connection of tackle, direction of the forces to obtain the maximum mechanical advantage.

(b) Calculate the M.A of this pulley system if its efficiency is 80%.



Question 6 **[3+3+4]**

(i) A ball of mass 20 g falls from a height of 10 m and after striking the ground, it rebounds from the ground to a height of 8 m. Calculate:

(a) the kinetic energy of the ball just before striking the ground.

(b) the loss in kinetic energy of the ball on striking the ground. [Take g =10 ms^{-2}]

(ii) A boy uses a single fixed pulley to lift a load of 50 kgf to some height. Another boy uses a single movable pulley to lift the same load to the same height.

(a) Compare the efforts applied by them showing necessary calculations.

(b) Show with necessary diagram how can one apply the effort in a convenient direction without altering the mechanical advantage of a single movable pully?

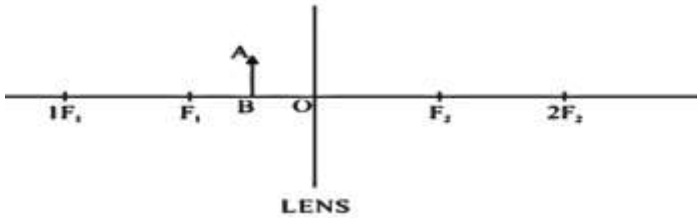
(iii) (a) When a tuning fork [vibrating] is held close to ear, one hears a faint hum. The same [vibrating tuning fork] is held such that its stem is in contact with the table surface, then one hears a loud sound. Explain.

(b) Why do the clouds appear white during the day?

(c) What is the acoustically measurable quantity related to pitch?

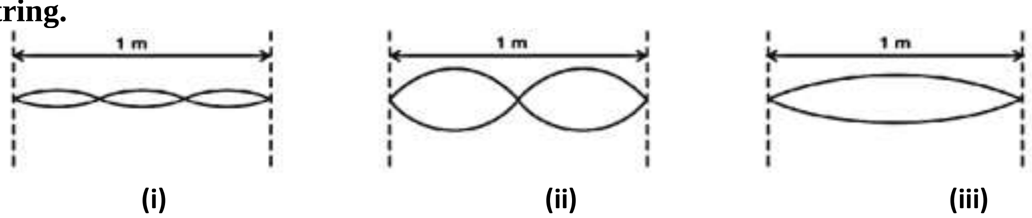
Question 7 **[4+3+3]**

(i) An object AB is placed between O and F_1 on the principal axis of a converging lens as shown in the diagram.



Copy the diagram and by using any two standard rays starting from point A, obtain an image of the object AB.

(ii) The diagrams given below show three modes of vibrations of a 1 m long stretched string.



(a) Which of the above modes of vibration will produce the shrillest sound?

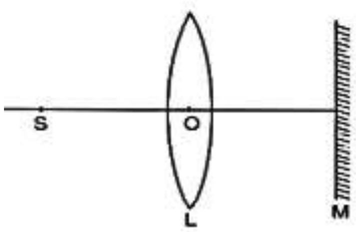
(b) If the frequency of the note produced in (i) is f, write down the frequency in cases (ii) and (iii).

(iii) The diagram shows a convex lens L, a plane mirror M and a point source of light S. Rays of light from the source S return to their point of origin.

(a) Complete the ray diagram to show this.

(b) What is the purpose of the plane mirror M?

(c) What is the effect of moving the mirror M towards the lens L on the return of rays of light?



Question 8 **[3+3+4]**

(i) a) A radiation X is focused by a particular device, on the bulb of a thermometer and mercury in the thermometer shows a rapid increase. Identify X.

b) Write any one use of X.

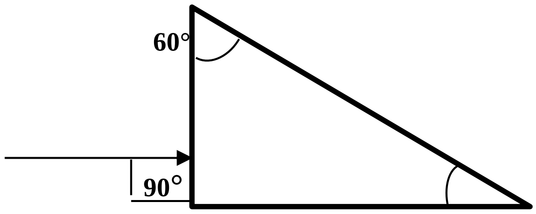
c) Mention a source of X.

(ii) a) With reference to the direction of force, how does a centripetal force differ from a centrifugal force during uniform circular motion?
b) Is centripetal force the force of reaction of centripetal force?
c) Compare the magnitudes of centripetal and centrifugal force.

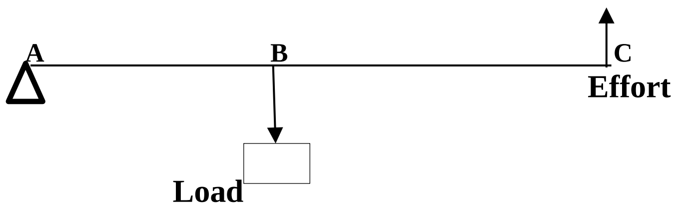
(iii) An electric motor of power 200 W is switched on for 1 min 40 s. If 60% of the energy is useful, calculate
a) useful work done by the motor.
b) weight lifted by it through a vertical height of 10m. [g = 10 ms^{-2}]
c) Mention the energy change in a bio-gas burner while in use.

Question 9 **[3+3+4]**

(i) Copy and complete the following diagram to show the path of the ray of monochromatic light as it enters and emerges out of the right-angled prism. Take critical angle of glass as 42°.



(ii) The diagram of a lever is given below. AB = 40 cm and BC = 60 cm.
a) To which class does it belong?
b) Calculate the mechanical advantage of the lever.
c) Calculate the effort required to lift a load of 75N.



(iii) a) Define dispersion.
b) Can dispersion take place through a hollow prism? Explain the reason in support of your answer.
c) What is the subjective property of light related to its wavelength?